

FSUB5100

## iso-Butyl alcohol

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 2-甲基-1-丙醇  
 iso-Butyl alcohol

**Cat No. :** B/5100/25, B/5100/27, B/5100/17, B/5100/PB17  
**Synonyms** Isobutanol; Isobutyl alcohol  
**CAS No** 78-83-1  
**Molecular Formula** C4 H10 O

**Supplier**
**UK entity/business name**  
 Fisher Scientific UK  
 Bishop Meadow Road, Loughborough,  
 Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
 Thermo Fisher Scientific  
 Janssen Pharmaceuticaaan 3a  
 2440 Geel, Belgium

**Emergency Telephone Number** Tel: 01509 231166  
 Chemtrec US: (800) 424-9300  
 Chemtrec EU: 001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
aromatic

#### Emergency Overview

Flammable liquid and vapor. Causes skin irritation. Causes serious eye damage. May cause respiratory irritation. May cause drowsiness and dizziness. May be harmful if swallowed. May be harmful in contact with skin.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Flammable liquids.                                 | Category 3 |
| Acute Oral Toxicity                                | Category 5 |
| Acute Dermal Toxicity                              | Category 5 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 1 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements

**Signal Word****Danger****Hazard Statements**

H226 - Flammable liquid and vapor  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H303 - May be harmful if swallowed  
H313 - May be harmful in contact with skin

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Vapors may cause flash fire or explosion. Flammable liquid.

**Health Hazards**

Causes skin irritation. Causes serious eye damage. May cause respiratory irritation. May cause drowsiness or dizziness. May be harmful if swallowed. May be harmful in contact with skin.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component        | CAS No  | Weight % |
|------------------|---------|----------|
| Isobutyl alcohol | 78-83-1 | 99       |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

None reasonably foreseeable. Causes severe eye damage. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Specific Hazards Arising from the Chemical**

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Storage

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component        | China | Taiwan                                    | Thailand | Hong Kong                                 |
|------------------|-------|-------------------------------------------|----------|-------------------------------------------|
| Isobutyl alcohol | -     | TWA: 50 ppm<br>TWA: 152 mg/m <sup>3</sup> |          | TWA: 50 ppm<br>TWA: 152 mg/m <sup>3</sup> |

| Component        | ACGIH TLV   | OSHA PEL                                                                                                    | NIOSH                                                       | The United Kingdom                                                                                               | European Union |
|------------------|-------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------|
| Isobutyl alcohol | TWA: 50 ppm | (Vacated) TWA: 50 ppm<br>(Vacated) TWA: 150 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 300 mg/m <sup>3</sup> | IDLH: 1600 ppm<br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | STEL: 75 ppm 15 min<br>STEL: 231 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 154 mg/m <sup>3</sup> 8 hr |                |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material  | Breakthrough time | Glove thickness | EU standard | Glove comments                                                                 |
|-----------------|-------------------|-----------------|-------------|--------------------------------------------------------------------------------|
| Nitrile rubber  | > 480 minutes     | 0.38 mm         | EN 374      | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Butyl rubber    | > 480 minutes     | 0.35 mm         | Level 6     |                                                                                |
| Neoprene gloves | > 480 minutes     | 0.45 mm         |             |                                                                                |

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|           |               |         |
|-----------|---------------|---------|
| Viton (R) | > 480 minutes | 0.70 mm |
|-----------|---------------|---------|

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Organic gases and vapours filter Type A Brown conforming to EN14387                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                                 |                                          |
|------------------------------------------------|-------------------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                                       |                                          |
| <b>Physical State</b>                          | Liquid                                          |                                          |
| <b>Odor</b>                                    | aromatic                                        |                                          |
| <b>Odor Threshold</b>                          | No data available                               |                                          |
| <b>pH</b>                                      | No information available                        |                                          |
| <b>Melting Point/Range</b>                     | -108 °C / -162.4 °F                             |                                          |
| <b>Softening Point</b>                         | No data available                               |                                          |
| <b>Boiling Point/Range</b>                     | 108 °C / 226.4 °F                               |                                          |
| <b>Flash Point</b>                             | 28 °C / 82.4 °F                                 | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | 0.6                                             | (Butyl Acetate = 1.0)                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                                  | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.6 Vol%<br><b>Upper</b> 10.9 Vol% |                                          |
| <b>Vapor Pressure</b>                          | 11.7 mbar @ 20°C                                |                                          |
| <b>Vapor Density</b>                           | 2.6                                             | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 0.800                                           |                                          |
| <b>Bulk Density</b>                            | Not applicable                                  | Liquid                                   |
| <b>Water Solubility</b>                        | Soluble                                         |                                          |
| <b>Solubility in other solvents</b>            | No information available                        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                  |                                          |
| Isobutyl alcohol                               | 1                                               |                                          |
| <b>Autoignition Temperature</b>                | 430 °C / 806 °F                                 |                                          |
| <b>Decomposition Temperature</b>               | No data available                               |                                          |
| <b>Viscosity</b>                               | No data available                               |                                          |
| <b>Explosive Properties</b>                    | Not explosive                                   | explosive air/vapour mixtures possible   |

**Oxidizing Properties** Not oxidising (based on the chemical structure of the substance and oxidation states of the constituent elements)

**Molecular Formula** C4 H10 O  
**Molecular Weight** 74.12  
**VOC Content(%)** 100 %

**SECTION 10. STABILITY AND REACTIVITY**

**Stability** Stable under normal conditions.

**Hazardous Reactions** None under normal processing.  
**Hazardous Polymerization** Hazardous polymerization does not occur.

**Conditions to Avoid** Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

**Materials to avoid** Strong oxidizing agents. Acid anhydrides. Acid chlorides.

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**SECTION 11. TOXICOLOGICAL INFORMATION**

**Product Information** See actual entry in RTECS for complete information.

**(a) acute toxicity;**

| Component        | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation               |
|------------------|---------------------------|------------------------------|-------------------------------|
| Isobutyl alcohol | LD50 = 2460 mg/kg ( Rat ) | LD50 = 3400 mg/kg ( Rabbit ) | LC50 > 18.18 mg/L ( Rat ) 6 h |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

**Respiratory** Based on available data, the classification criteria are not met  
**Skin** Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

**(h) STOT-single exposure;** Category 3

**Results / Target organs** Respiratory system  
Central nervous system (CNS)

**(i) STOT-repeated exposure;** Based on available data, the classification criteria are not met

**Target Organs** None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Do not empty into drains. .

| Component        | Freshwater Fish                                                                                                                                                                                                                                          | Water Flea                                                                                   | Freshwater Algae                              | Microtox                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------|
| Isobutyl alcohol | LC50: 1370 - 1670 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 375 mg/L, 96h static (Pimephales promelas)<br>LC50: 1120 - 1520 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 1480 - 1730 mg/L, 96h flow-through (Lepomis macrochirus) | EC50: = 1300 mg/L, 48h (Daphnia magna)<br>EC50: 1070 - 1933 mg/L, 48h Static (Daphnia magna) | 1799 mg/l EC50 = 72 h<br>230 mg/L EC50 = 48 h | EC50 = 1224.6 mg/L 15 min |

**Persistence and Degradability** Readily biodegradable  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.

| Component                          | Degradability |
|------------------------------------|---------------|
| Isobutyl alcohol<br>78-83-1 ( 99 ) | 90% (14d)     |

**Degradation in sewage treatment plant** Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Product has a low potential to bioconcentrate; Bioaccumulation is unlikely

| Component        | log Pow | Bioconcentration factor (BCF) |
|------------------|---------|-------------------------------|
| Isobutyl alcohol | 1       | < 100                         |

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Disperses rapidly in air: Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

UN-No UN1212  
Proper Shipping Name ISOBUTANOL  
Hazard Class 3  
Packing Group III

IMDG/IMO

UN-No UN1212  
Proper Shipping Name ISOBUTANOL  
Hazard Class 3  
Packing Group III

IATA

UN-No UN1212  
Proper Shipping Name ISOBUTANOL  
Hazard Class 3  
Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component        | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Isobutyl alcohol | X                                                   | X                                       | X    | X     | 201-148-0 | X    | X   | X     | X    | X    | X    | KE-24894 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 22-Jan-2009  
Revision Date 04-Apr-2024  
Revision Summary Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

First aid for chemical exposure, including the use of eye wash and safety showers.



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**Legend****CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**